FORMAL-SECTIONS ALGEBRAIZATION THEOREM: COHOMOLOGY OF F ON SMALL OPENS EQUALS THE FORMAL LIMIT BELOW DEGREE S, WITH FINITENESS UNDER A STRONGER DEPTH BOUND

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3768-3808 of sources/stacks-algebraization/pinned-original-source.tex; proof span is lines 3810-3846, the complete author proof (35 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool algebraization wave36 (al02 bib-salvage batch). Spans computed from \label (16.5/16.6).

Editorial note. External \cite keys salvaged with card-local bibliography.tex entries copied verbatim from the snapshot my.bib (precedent S3-morph-blow-up-ample-family); URL/ISSN/EPRINT fields dropped as the preamble has no url package.

Editorial note. statement contains reference environment with real content -> patch-sd-envs.tex applied (sd02 lesson).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; in-chapter lemmas cited by name become stub prereqs; external cites resolved via the card-local bibliography.tex fragment.

Source reference. Original theorem 0DXP: sources/stacks-algebraization/pinned-original-source.tex, source lines 2720–2735. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0DXI: sources/stacks-algebraization/pinned-original-source.tex, source lines 3736–3746. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0DXN: sources/stacks-algebraization/pinned-original-source.tex, source lines 2688–2694. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0BK0: sources/stacks-algebraization/pinned-original-source.tex, source lines 3846–3846. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Theorem 0.1. *The method of proof follows roughly the method of proof of* [1, Theorem 1] *and* [2, Satz 2]. *The result is almost the same as* [4, Theorem 1.1] *(affine complement case) and* [3, Theorem 3.9] *(complement is union of few affines). Let $(A, \mathfrak{m})$ be a Noetherian local ring which has a dualizing complex and is complete with respect to an ideal I. Set $X = \mathrm{Spec}(A)$, $Y = V(I)$, and $U = X \setminus \{\mathfrak{m}\}$. Let $\mathcal{F}$ be a coherent sheaf on U. Assume*

(1) *$cd(A, I) \leq d$, i.e., $H^i(X \setminus Y, \mathcal{G}) = 0$ for $i \geq d$ and quasi-coherent $\mathcal{G}$ on X,*
(2) *for any $x \in X \setminus Y$ whose closure $\overline{\{x\}}$ in X meets $U \cap Y$ we have*

$$depth_{\mathcal{O}_{X,x}}(\mathcal{F}_x) \geq s \quad or \quad depth_{\mathcal{O}_{X,x}}(\mathcal{F}_x) + \dim(\overline{\{x\}}) > d + s$$

Then there exists an open $V_0 \subset U$ containing $U \cap Y$ such that for any open $V \subset V_0$ containing $U \cap Y$ the map

$$H^i(V, \mathcal{F}) \to \lim H^i(U, \mathcal{F}/I^n\mathcal{F})$$

is an isomorphism for $i < s$. If in addition $depth_{\mathcal{O}_{X,x}}(\mathcal{F}_x) + \dim(\overline{\{x\}}) > s$ for all $x \in U \cap Y$, then these cohomology groups are finite A-modules.

Proof. Choose a finite A-module M such that $\mathcal{F}$ is the restriction to U of the coherent $\mathcal{O}_X$-module associated to M, see Local Cohomology, Lemma 0BK0. Then the assumptions of Lemma 0DXP are satisfied. Pick J_0 as in that lemma and set $V_0 = X \setminus V(J_0)$. Then opens $V \subset V_0$ containing $U \cap Y$ correspond 1-to-1 with ideals $J \subset J_0$ with $V(J) \cap V(I) = \{\mathfrak{m}\}$. Moreover, for such a choice we have a distinguished triangle

$$R\Gamma_J(M) \to M \to R\Gamma(V, \mathcal{F}) \to R\Gamma_J(M)[1]$$

We similarly have a distinguished triangle

$$R\Gamma_{\mathfrak{m}}(M)^\wedge \to M \to R\Gamma(U, \mathcal{F})^\wedge \to R\Gamma_{\mathfrak{m}}(M)^\wedge[1]$$

involving derived I-adic completions. The cohomology groups of $R\Gamma(U, \mathcal{F})^\wedge$ are equal to the limits in the statement of the theorem by Lemma 0DXI. The canonical map between these triangles and some easy arguments show that our theorem follows from the main Lemma 0DXP (note that we have $i < s$ here whereas we have $i \leq s$ in the lemma; this is because of the shift). The finiteness of the cohomology groups (under the additional assumption) follows from Lemma 0DXN. □

References

[1] Faltings, Gerd. *Algebraisation of some formal vector bundles*. Ann. of Math. (2) 110 (3), 501–514 (1979).

[2] Faltings, Gerd. *Über lokale Kohomologiegruppen hoher Ordnung*. J. Reine Angew. Math. 313, 43–51 (1980).

[3] Raynaud, Michèle. *Théorèmes de Lefschetz en cohomologie cohérente et en cohomologie étale*. Société Mathématique de France, Paris, 1975. Bull. Soc. Math. France, Mém. No. 41, Supplément au Bull. Soc. Math. France, Tome 103.

[4] Raynaud, Michèle. *Théoremes de Lefschetz en cohomologie des faisceaux cohérents et en cohomologie étale. Application au groupe fondamental*. Ann. Sci. École Norm. Sup. (4) 7, 29–52 (1974). Collection of articles dedicated to Henri Cartan on the occasion of his 70th birthday, I.